

THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2021

Mathematics

MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2

(2019–2020 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes**Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2

(Multiple Choice Questions for SDE Candidates)

1. $\frac{d}{dn}(\ln kx) = \underline{\hspace{2cm}}.$

(A) $kx.$

(B) $\frac{1}{kx}.$

(C) $\frac{1}{x}.$

(D) $\frac{k}{\ln x}.$

2. $\sinh x = \frac{-3}{4}$. Then $\cosh x$ is $\underline{\hspace{2cm}}.$

(A) $\frac{1}{4}.$

(B) $\frac{5}{4}.$

(C) $\frac{4}{5}.$

(D) $\frac{-3}{5}.$

3. $\frac{d}{dx} \cosh^{-1}(x^2) = \underline{\hspace{2cm}}.$

(A) $2x \sinh^{-1}(x^2).$

(B) $\frac{2x}{\sqrt{x^2-1}}.$

(C) $\frac{x}{x^4-1}.$

(D) $\frac{2x}{\sqrt{x^4-1}}.$

4. $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \underline{\hspace{2cm}}.$

(A) $\ln\left(\frac{a}{b}\right).$

(B) $\ln\left(\frac{b}{a}\right).$

(C) $\ln(ab).$

(D) $\infty.$

5. $\lim_{x \rightarrow 0} x \cot x = \underline{\hspace{2cm}}.$

(A) 0.

(B) 1.

(C) $x.$

(D) -1.

6. $a_1 = 2, a_{n+1} = a_n + 3$. Then $a_2 = \underline{\hspace{2cm}}.$

(A) 5.

(B) 3.

(C) 6.

(D) 7.

7. $a_n = \frac{2n+1}{3n+5}$. The $\lim_{n \rightarrow \infty} a_n =$ _____.

(A) $\frac{1}{5}$.

(B) $\frac{2}{5}$.

(C) $\frac{2}{3}$.

(D) $\frac{3}{2}$.

8. $\lim_{n \rightarrow \infty} \frac{3^n}{n^2} =$ _____.

(A) 3.

(B) $\frac{3}{2}$.

(C) 1.

(D) ∞ .

9. If $|r| > 1$, $\lim_{n \rightarrow \infty} r^n =$ _____.

(A) 0.

(B) 1.

(C) ∞ .

(D) None of these.

10. Let $\{a_n\}$ of $\{b_n\}$ is such that $a_n \leq b_n$. Then :

(A) $\sum a_n$ converges if $\sum b_n$ converges. (B) $\sum b_n$ converges if $\sum a_n$ converges.

(C) $\sum a_n$ converges if $\sum b_n$ diverges. (D) $\sum b_n$ diverges if $\sum a_n$ diverges.

11. $\sum_{n=1}^{\infty} \frac{n}{n^3+1}$ is:

(A) Converges to 1.

(B) Converges to 0.

(C) Diverges.

(D) None of these.

12. If a series converges absolutely for all x . Then its radius of converges if :

(A) Finite.

(B) Infinite.

(C) Cannot be determined.

(D) None of these.

13. The series $\sum_{n=0}^{\infty} x^n$ is :

(A) Converges absolutely for $|x| < 1$. (B) Converges for $|x| > 1$.

(C) Has radius of converges $\frac{1}{2}$. (D) None of these.

Turn over

14. Find $\frac{d^2y}{dx^2}$ if $x = \cos t, y = \sin t$:
- (A) $\operatorname{cosec}^2 t$. (B) $-\operatorname{cosec}^2 t$.
 (C) $\operatorname{cosec}^3 t$. (D) $-\operatorname{cosec}^3 t$.
15. The equation $r^2 = \sin 2\theta$ is symmetric about :
- (A) x -axis. (B) y -axis.
 (C) Origin. (D) The line $\theta = \frac{\pi}{4}$.
16. Polar equation of the circle with centre at $\left(-2, \frac{\pi}{2}\right)$ and passing through origin is :
- (A) $r = -4 \sin \theta$. (B) $r = 4 \sin \theta$.
 (C) $r = 4 \cos \theta$. (D) $r = -4 \cos \theta$.
17. $\ln(e^x) = \underline{\hspace{2cm}}$.
- (A) e^x . (B) x .
 (C) $\frac{1}{e^x}$. (D) $\frac{1}{x}$.
18. $\sum (-1)^n \frac{1}{n}$ is $\underline{\hspace{2cm}}$.
- (A) Converges. (B) Diverges.
 (C) Absolutely convergent. (D) None.
19. $y = \operatorname{sech}^2 x + \tanh^2 x$ then $\frac{dy}{dx} = \underline{\hspace{2cm}}$.
- (A) $-2 \operatorname{sech} x \tanh x$. (B) 0.
 (C) 1. (D) $\operatorname{sech}^2 x \tanh x$.
20. The series $\sum n^m x^n$ is converged if $\underline{\hspace{2cm}}$.
- (A) $x > 1$ and $x = 1$ when $m < -1$. (B) $x > 1$ and $x = 1$ when $m > -1$.
 (C) $x < 1$ and $x = 1$ when $m < -1$. (D) $x < 1$ and $x = 1$ when $m > -1$.

THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2021

Mathematics

MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2

(2019–2020 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Determine whether the function $f(x) = x^3 - 3x + 1$ has an inverse.
2. Find the derivatives of (a) $3^{\sqrt{x}}$; (b) $\cos^{-1}(3x)$.
3. Find the derivative of $\log \left[\frac{x^2(2x^2+1)^3}{\sqrt{5-x^2}} \right]$ when $x = 1$.
4. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$.
5. Find $\lim_{x \rightarrow \infty} \frac{\log n}{n}$.
6. Determine whether the series converges. If it converges find the sum $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$.
7. Use integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ converges or diverges.
8. Show that the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges.

Turn over

9. Find the Maclaurin's series of $f(x) = \cos x$.
10. Find the radius of convergence and interval of convergence of the power series $\sum_{n=0}^{\infty} n! x^n$.
11. Describe the curve represented by $x = 4 \cos \theta$ and $y = 3 \sin \theta$, $0 \leq \theta \leq 2\pi$.
12. Find the angle between the two planes defined by $3x - y + 2z = 1$ and $2x + 3y - z = 4$.
13. Find an equation in rectangular co-ordinates for the surface with the cylindrical co-ordinates $r^3 \cos 2\theta - z^2 = 4$.
14. Find a vector function that describes the curve of intersection of the cylinder $x^2 + y^2 = 4$ and the plane $x + y + 2z = 4$.
15. Evaluate $\int_0^1 r(t) dt$ if $r(t) = t^2 i + \frac{1}{t+1} j + e^{-t} k$.

(10 × 3 = 30 marks)

Section B

*Answer at least five questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 30.*

16. Use logarithmic differentiation to find the derivative of $y = \sqrt[3]{\frac{x-1}{x^2+1}}$.
17. Find the derivative of $y = x^2 \operatorname{sech}^{-1}(3x)$.
18. Evaluate $\int_0^1 \log x dx$.
19. Show that the series $\sum_{n=1}^{\infty} \left[\frac{2}{n(n+1)} - \frac{4}{3^n} \right]$ is convergent and find its sum.
20. Find the tangent lines of $r = \cos 2\theta$ at the origin.

21. Find the length of the Cardioid $r = 1 + \cos \theta$.
22. Find the parametric equations for the line of intersection of the planes defined by $3x - y + 2z = 1$ and $2x + 3y - z = 4$.
23. Find the velocity vector, acceleration vector and speed of a particle with position vector :
- $$r(t) = \sqrt{t} \, i + tt^2 \, j + e^{2t} \, k, t \geq 0.$$

(5 × 6 = 30 marks)

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Find the derivative of $\sec^{-1}(e^{-2x})$.
- (b) Evaluate $\lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\sin x}$
25. (a) Find the area S of the surface obtained by revolving the circle $r = \cos \theta$ about the line $\theta = \pi/2$.
- (b) Show that the surface area of a sphere of radius r is $4\pi r^2$.
26. (a) Determine whether the series $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ is convergent or divergent.
- (b) Show that sequence $\left\{\frac{2^n}{n!}\right\}$ is convergent and find its limit.
27. (a) Find an equation in rectangular co-ordinates for the surface with spherical equation $\rho = 4 \cos \phi$.
- (b) A moving object has an initial position and an initial velocity given by the vectors $r(0) = i + 2j + k$ and $v(0) = i + 2k$. Its acceleration at time t is $a(t) = 6t \, i + j + 2k$. Find its velocity and position at time t .

(2 × 10 = 20 marks)